# Siddhant Kotak

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## Professional Experience

**Techuz InfoWeb** — ***TypeScript, Node.js, React, Azure, PostgreSQL*** **Mar 2026 – Present**

*Fullstack + AI Engineer* *Ahmedabad, Gujarat*

### *SkyConnect: HIPAA-Compliant Multi-Tenant Clinical Messaging Platform*

- Designed the platform as one container image with three entry points: a stateless REST API, a Socket.IO gateway with a Redis adapter for multi-replica fan-out, and Service Bus workers, deployed as three independently scaled Azure Container Apps.
- Enforced tenant isolation with PostgreSQL FORCE row-level security on **34 tables**, setting the tenant GUC via `SET LOCAL` inside a transaction because session-scoped GUCs leak between requests under pgBouncer transaction pooling.
- Moved push dispatch, thumbnailing, video transcoding and link-preview fetching off the request path onto **13 publish/subscribe topics** with dead-letter queues, bounded retries and idempotent consumers.
- Guaranteed ordering and exactly-once sends with a server-assigned per-conversation sequence and client idempotency keys, so a retried send returns the original row and no conflict-resolution service was needed.
- Scaled the worker tier with database leases electing a single runner per scheduled job, and hardened Azure: VNet-integrated Container Apps, Redis private endpoint, Key Vault via managed identity, digest-verified image promotion.
- Built HIPAA controls into the schema: Entra ID SSO with opaque hashed sessions and per-request revocation, zero-PHI push, 30-minute SAS URLs so the API never proxies protected bytes, and an append-only partitioned audit log that rejects UPDATE and DELETE. Covered by **627 backend and 686 frontend tests**.

**Demaze Technologies** — ***Python, FastAPI, ML/AI, MERN*** **Sept 2025 – Feb 2026**

*ML/AI + Fullstack Engineer* *Ahmedabad, Gujarat*

### *AI-Powered Fashion E-Commerce Platform (DesignerKidsClub)*

- Across 15+ FastAPI microservices, engineered a multi-modal search system with an 8-tier ranking algorithm, CLIP-based visual search and semantic search over Sentence Transformers, served from Elasticsearch 8.x KNN vector search at 85%+ accuracy with sub-second responses.
- Built recommendation engines with multi-factor scoring (35% semantic, 20% brand, 18% category), seasonal weather integration and LLM-powered trend forecasting.
- Implemented the GPT-4o vision API for automated wardrobe feature extraction and OpenCLIP 768-dimension embeddings, with AI-generated tagging across 1000+ products.

### *Trading Risk and Market Sentiment Platform (RMS)*

- Architected a three-engine ML ensemble (40% Behavioral XGBoost, 35% Leverage Random Forest, 25% Volatility XGBoost) analysing 146+ engineered features at 85–92% accuracy, served from a FastAPI backend at sub-10ms health checks and 2–5s inference.
- Built the companion sentiment platform monitoring 30+ premium sources (Reuters, Bloomberg, Economic Times), processing 19,200+ articles daily at 99.9% uptime via mirror-feed fallback and circuit-breaker protection.
- Automated the MLOps pipeline with Celery-based weekly retraining, Optuna Bayesian optimisation and MLflow experiment tracking.
- Engineered dual-model sentiment analysis integrating FinBERT (85% accuracy on financial text) and Twitter-RoBERTa with batch processing, LRU caching and a compound scoring algorithm.
- Implemented an explainability layer combining SHAP for global feature importance, LIME for individual interpretations and GPT-4 for executive-level natural-language risk summaries with counterfactual analysis.

## Personal Projects

**Shizuku: AI-Powered Digital Studio for Manga and Storytelling** **Ongoing**

*React 18, Express.js, TypeScript, Drizzle ORM, Redis* *Personal Project*

- Building an AI-driven digital studio for manga adaptation, document processing and audio synthesis, structured as a Turborepo monorepo sharing 16+ internal packages across applications.
- Architected a custom 2.5D spatial engine mapping characters to grid coordinates to hold camera perspective and scene continuity consistent across multi-panel layouts.
- Engineered multi-modal generation pipelines combining text-to-layout synthesis, panel generation and voice synthesis, with automated hash-based copyright safeguards.
- Backed it with type-safe relational schemas over PostgreSQL and Drizzle ORM, paired with Redis for low-latency session management and pub/sub messaging.

## Technical Skills

**Languages**: TypeScript, Python, JavaScript, SQL, Java

**Systems Design**: publish/subscribe with competing consumers, dead-letter queues and bounded retries, idempotent consumers, lease-based leader election, multi-tenant row-level security, exactly-once delivery, HIPAA-compliant architecture

**Backend and Realtime**: Node.js, Express 5, FastAPI, Socket.IO, REST API design, Knex, Zod, Celery

**Data**: PostgreSQL (row-level security, partitioning, query tuning), Redis, MongoDB, Elasticsearch KNN vector search, Pandas, NumPy

**Cloud and Delivery**: Azure (Container Apps, Service Bus, Blob Storage, Key Vault, Entra ID, API Management), AWS, Docker, Turborepo, digest-verified image promotion

**ML and AI**: XGBoost, Scikit-learn, PyTorch, model ensembling, MLOps (MLflow, Optuna, scheduled retraining), explainability (SHAP, LIME), embeddings and semantic retrieval (OpenCLIP, Sentence Transformers, FinBERT)

**Frontend and Testing**: React 19, TanStack Query, TailwindCSS, Vite, Pytest, Vitest

## Extracurricular

**Tic-Tech-Toe'24 Hackathon** **2024**

*Shortlisted top 14 of 200+ teams, 48-hour national-level hackathon* *DAIICT*

- Led full-stack development of a React.js and Node.js application with database integration, RESTful APIs and modern deployment pipelines.
- Project: github.com/SiddhantKotak/Tic-Tech-Toe-24

## Education

**Lok Jagruti Institute of Engineering and Technology** **June 2026**

*Computer Science Engineering* *Ahmedabad, Gujarat*